

Tutorial 1**References**

- Topic 1: Logic
- Sections 2.1, 2.2, 2.4 of Epp (4th or 5th Ed.)

Work on the following problems in groups during the tutorial in week 2.

1. [B] Construct a truth table for each of the following compound statements, where p and q denote primitive statements.

(a) $(p \wedge (p \rightarrow q)) \rightarrow q$

(b) $q \leftrightarrow (\neg p \vee \neg q)$

Solution:

- (a) The truth table for $(p \wedge (p \rightarrow q)) \rightarrow q$ is given below.

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$(p \wedge (p \rightarrow q)) \rightarrow q$
0	0	1	0	1
0	1	1	0	1
1	0	0	0	1
1	1	1	1	1

- (b) The truth table for $q \leftrightarrow (\neg p \vee \neg q)$ is given below.

p	q	$\neg p$	$\neg q$	$\neg p \vee \neg q$	$q \leftrightarrow (\neg p \vee \neg q)$
0	0	1	1	1	0
0	1	1	0	1	1
1	0	0	1	1	0
1	1	0	0	0	0

2. [M] Determine all truth values assignments, if any, for the primitive statements p, q, r, s, t that make the following compound statement false:

$$((p \wedge q) \wedge r) \rightarrow (s \vee t).$$

Solution:

The compound statement $((p \wedge q) \wedge r) \rightarrow (s \vee t)$ is false (0) when $(p \wedge q) \wedge r$ is true (1) and $(s \vee t)$ is false (0). Hence p, q , and r must be true (1) while s and t must be false (0). (Draw the truth table to see this more clearly.)

3. [M] Let p, q, r be primitive statements. Use truth tables to verify the following logical equivalence:

$$((p \vee q) \rightarrow r) \Leftrightarrow ((p \rightarrow r) \wedge (q \rightarrow r)).$$

Solution:

p	q	r	$p \vee q$	$(p \vee q) \rightarrow r$	$p \rightarrow r$	$q \rightarrow r$	$(p \rightarrow r) \wedge (q \rightarrow r)$
0	0	0	0	1	1	1	1
0	0	1	0	1	1	1	1
0	1	0	1	0	1	0	0
0	1	1	1	1	1	1	1
1	0	0	1	0	0	1	0
1	0	1	1	1	1	1	1
1	1	0	1	0	0	0	0
1	1	1	1	1	1	1	1

Since the truth tables of $(p \vee q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are the same (see the columns in bold), it follows that $((p \vee q) \rightarrow r) \Leftrightarrow ((p \rightarrow r) \wedge (q \rightarrow r))$.

- $\Rightarrow$ 4. [M] Determine whether the following statements are logically equivalent. In each case, construct a truth table, and explain your answer.

(a) $p \vee (p \wedge q)$ and p .

(b) $(p \wedge q) \vee r$ and $p \wedge (q \vee r)$.

Solution:

(a)

p	q	$p \wedge q$	$p \vee (p \wedge q)$
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1

Since the bold columns are equivalent, these are logically equivalent statements.

(b)

p	q	r	$p \wedge q$	$(p \wedge q) \vee r$	$q \vee r$	$p \wedge (q \vee r)$
0	0	0	0	0	0	0
0	0	1	0	1	1	0
0	1	0	0	0	1	0
0	1	1	0	1	1	0
1	0	0	0	0	0	0
1	0	1	0	1	1	1
1	1	0	1	1	1	1
1	1	1	1	1	1	1

Since the bold columns are not equivalent, neither are the statements.

5. [B] Use truth tables to establish which of these statements are tautologies, and which are contradictions.

(a) $(p \wedge q) \vee (\neg p \vee (p \wedge \neg q))$

(b) $(p \wedge \neg q) \wedge (\neg p \vee q)$

(c) $(\neg p \vee q) \vee (p \wedge \neg q)$

Solution:

(a)

p	q	$p \wedge q$	$p \wedge \neg q$	$\neg p \vee (p \wedge \neg q)$	$(p \wedge q) \vee (\neg p \vee (p \wedge \neg q))$
0	0	0	0	1	1
0	1	0	0	1	1
1	0	0	1	1	1
1	1	1	0	0	1

The bold column indicates this is a tautology.

(b)

p	q	$p \wedge \neg q$	$\neg p \vee q$	$(p \wedge \neg q) \wedge (\neg p \vee q)$
0	0	0	1	0
0	1	0	1	0
1	0	1	0	0
1	1	0	1	0

The bold column indicates this is a contradiction.

(c)

p	q	$\neg p \vee q$	$p \wedge \neg q$	$(\neg p \vee q) \vee (p \wedge \neg q)$
0	0	1	0	1
0	1	1	0	1
1	0	0	1	1
1	1	1	0	1

The bold column indicates this is a tautology.

6. [B] Write each of these statements in symbolic form, and determine whether they are logically equivalent. Include a truth table and explanation in your answer.

If you paid full price, you didn't buy it at Crown Books.

You didn't buy it at Crown Books or you paid full price.

Solution:

Using p for *you paid full price* and c for *you bought it at Crown Books*, we write:

p	c	$p \rightarrow \neg c$	$\neg c \vee p$
0	0	1	1
0	1	1	0
1	0	1	1
1	1	0	1

The bold columns are not the same, indicating the statements are not logically equivalent.

⇒ 7. [M] Write a negation of the following statements.

(a) If n is prime, then n is odd or n is 2.

(b) If n is divisible by 6, then n is divisible by 2 and n is divisible by 3.

Solution:

(a) Using p for n is prime, q for n is odd, and r for n is 2:

Given $p \rightarrow (q \vee r)$, the negation is $p \wedge \neg(q \vee r)$, or equivalently $p \wedge (\neg q \wedge \neg r)$.

This could be expressed as n is an even prime other than 2, though other wordings are possible.

(b) Using 6 for n is divisible by 6, and similarly for 2 and 3:

Given $6 \rightarrow (2 \wedge 3)$, the negation is $6 \wedge \neg(2 \wedge 3)$, or equivalently $6 \wedge (\neg 2 \vee \neg 3)$.

This could be expressed as n is divisible by 6, and either odd or not divisible by 3, although other wordings are possible.

8. [M] Use $\leftrightarrow$ to convert this logical equivalence to a tautology. Then use a truth table to verify it is a tautology.

$$p \rightarrow (q \vee r) \equiv (p \wedge \neg q) \rightarrow r$$

Solution:

The tautology is $(p \rightarrow (q \vee r)) \leftrightarrow ((p \wedge \neg q) \rightarrow r)$, which has the following truth table. The bold columns being equivalent verifies the statement as a tautology.

p	q	r	$q \vee r$	$p \rightarrow (q \vee r)$	$p \wedge \neg q$	$(p \wedge \neg q) \rightarrow r$
0	0	0	0	1	0	1
0	0	1	1	1	0	1
0	1	0	1	1	0	1
0	1	1	1	1	0	1
1	0	0	0	0	1	0
1	0	1	1	1	1	1
1	1	0	1	1	0	1
1	1	1	1	1	0	1

9. [M] Consider the NAND statement form: $p \bar{\wedge} q$, which is equivalent to $\neg(p \wedge q)$.

For each of the following, construct a truth table, and identify which single logical connective could replace the NAND form.

(a) $\neg p \bar{\wedge} \neg q$

(b) $p \bar{\wedge} \neg q$

Solution:

Consider the following truth table:

p	q	$p \bar{\wedge} q$	$\neg p$	$\neg q$	$\neg p \bar{\wedge} \neg q$	$p \bar{\wedge} \neg q$
0	0	1	1	1	0	1
0	1	1	1	0	1	1
1	0	1	0	1	1	0
1	1	0	0	0	1	1

- (a) The truth table for $\neg p \bar{\wedge} \neg q$ is equivalent to that of $p \vee q$.
- (b) The truth table for $p \bar{\wedge} \neg q$ is equivalent to that of $p \rightarrow q$.